

ECCT 2026: Responses to the Reviewers' Queries/Comments

Paper ID: 139

Paper Title: Predicting Student Academic Performance from Collaborative Learning Behaviors Using Machine Learning and Explainable Insights

Author(s): First Author, Second Author, Third Author

Reviewer 1

Concern 5: SMOTE applied without discussion of impact.

Author(s) response: Thank you for this observation. We have added a dedicated subsection explaining why SMOTE was used, how it works, and how it was applied (exclusively to the training partition) to prevent data leakage and preserve evaluation integrity.

Author(s) action in the paper: Section 3.5 'Class Balancing with SMOTE' has been added to the revised paper, providing a full explanation of the rationale, mechanism, and controlled application of SMOTE.

Concern 1–4, 6–10:

Author(s) response: We acknowledge the reviewer's concerns regarding survey bias, binary labeling, potential overfitting, lack of cross-validation, feature subjectivity, absence of external validation, limited novelty, explainability depth, and causal interpretation. These points are important for strengthening the study and have been considered carefully.

Author(s) action in the paper: These aspects have been incorporated as future research directions. The Conclusion section has been extended to include future work on improving data sources (e.g., combining survey with objective data), adopting multi-class prediction, applying cross-validation techniques, validating on external datasets, enhancing explainability analysis, refining feature design, and avoiding causal interpretation without proper justification.

Reviewer 2

Concern 1: Lack of comparison with previous works. Authors need to clearly explain how their study is effective compared to existing methods.

Author(s) response: We thank the reviewer for this important observation. A new subsection (Section 2.1) has been added to compare this study with closely related prior work. The comparison highlights differences in data types, methodological approaches, and study objectives. In particular, prior studies primarily focus on explanatory or statistical analysis of collaborative learning and academic outcomes, whereas the present work adopts a machine learning-based predictive modeling framework.

Author(s) action in the paper: Section 2.1 "Comparison with Prior Predictive Work" has been added to the revised manuscript, including a comparison table (Table 1) that contrasts this study with selected prior works and clearly positions the contribution of the proposed predictive approach.

Concern 2: Figure 1 should be improved.

Author(s) response: We have analysed the Figure 1 and redrawn it.

Author(s) action in the paper: Figure 1 has been improved with relevance.

Concern 3: No relevant papers discussing similar issues from the perspective of local students were found, which is unlikely.

Author(s) response: We have reviewed some papers and discussed the closely related papers .

Author(s) action in the paper: We have updated literature review with the closely related paper.

Reviewer 3

Concern 1: Random conversion of categorical variables into numeric values is a major methodological flaw. Assigning arbitrary numbers to nominal data (like 'Study Location') imposes false ordinal relationships. One-hot encoding or label encoding should be used.

Author(s) response: we have studied the flaw and take the proper action to use one hot encoding.

Author(s) action in the paper: We have improvised the analysis with one hot encoding

Concern 2: Categorizing responses into Low (1), Medium (2), and High (3) preserves logical relationships and allows models to interpret magnitude correctly.

Author(s) response: Categorizing responses into Low (1), Medium (2), and High (3) preserves logical relationships and allows models to interpret magnitude correctly.

Author(s) action in the paper: To reflect this, we have explicitly clarified in the revised manuscript that the sentiment labels are treated as ordinal categories, preserving their natural ranking. We also added a brief explanation in the Data Annotation / Preprocessing section highlighting that this encoding facilitates better learning of sentiment intensity by the model

Concern 3: The manuscript does not explain why these specific 13 features were selected or how they are theoretically related to student performance. The inclusion of variables and their pedagogical relevance must be justified.

Author(s) response: This is a very valid concern. We have added a dedicated section that provides a thorough theoretical and empirical justification for each of the 13 features included in the study. Each feature is linked to relevant literature on collaborative learning, motivation, and academic performance.

Author(s) action in the paper: Section 3.3 'Feature Selection Rationale' has been added to the revised manuscript. It provides a literature-grounded justification for each feature, explaining its theoretical relevance to student academic performance within collaborative learning contexts.

Concern 4: The AI-generated content score is 53%. The manuscript must be manually rewritten to ensure original scholarly authorship.

Author(s) response: We acknowledge this concern and have substantially rewritten the manuscript to ensure the text reflects original scholarly authorship. All sections including the Abstract, Introduction, Literature Review, Methodology, Results, and Conclusion have been revised using the authors' own academic writing to reduce AI-generated content.

Author(s) action in the paper: The entire manuscript has been manually rewritten. AI-detected passages (flagged by Turnitin at 53%) have been replaced with original author-written text. The revised version is expected to show a significantly lower AI content score.

Concern 5: Accuracy of 0.977 is unusually high for survey data. Re-run after correct encoding technique; ensure results are not inflated by data leakage.

Author(s) response: We thank the reviewer for raising this important concern. We acknowledge that an accuracy of 0.977 may appear unusually high for survey-based sentiment data and warrants careful verification. To address this, we thoroughly re-examined our preprocessing and evaluation pipeline to ensure that no data leakage or improper encoding influenced the results

Author(s) action in the paper: We verified that label encoding was applied only after the train–test split, ensuring that no information from the test set was used during training